Case Study 2

Truss Joint Force Analysis

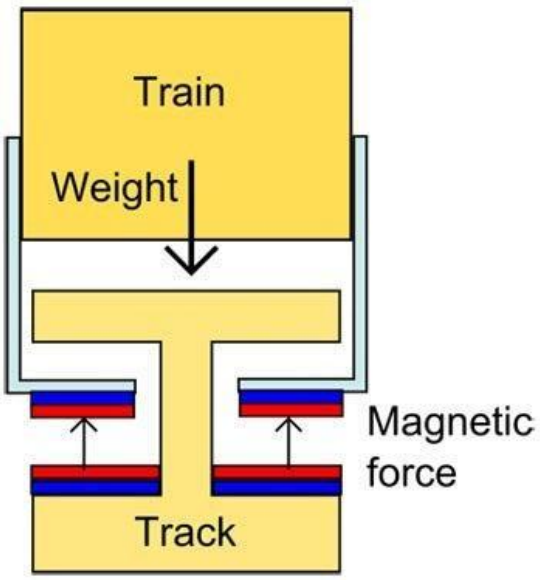
Assignment: A section of a maglev train of mass $m$is levitated using an electromagnetic suspension system. The magnetic force generated by the electromagnet is proportional to the square of the current and inversely proportional to the square of the air gap between the train and the track.

The nonlinear equation governing the vertical motion of the train is:

$$m\ddot{z} = mg - \frac{ki^2}{z^2}$$

where:

- $z(t)$is the levitation gap
- $i(t)$is the coil current
- $k$is a magnetic constant



Let's assume,

z = air gap between electromagnet & rail
i = current through electromagnet
m = mass supported by one electromagnet
g = gravitational acceleration
F_m = magnetic attractive force

Weight acting downwards : $W = mg$
Upward Magnetic force : $F_m = \frac{ki^2}{z^2}$ k = magnetic constant

We know, $\Sigma F = mz$

$$mg - F_m = mz$$

$$mz = mg - F_m$$

$$\therefore \quad mz = mg - \frac{ki^2}{z^2}$$

At equilibrium, $z = 0$

$$\therefore \quad 0 = mg - \frac{ki^2}{z^2}$$

$$\therefore mg = \frac{ki^2}{z^2} \qquad \therefore \quad \boxed{ki^2 = mgz^2}$$ Equilibrium Equation

When small disturbance is introduced

$z(t) = z_0 + x(t)$ where z_0 = equilibrium gap
$x(t)$ = small displ from equilbm.

$i(t) = i_0 + u(t)$ $u(t)$ = small current variation

Substitute this in $\left(mz = mg - \frac{ki^2}{z^2}\right)$

$$\therefore \quad mx = mg - \frac{k(i_0+u)^2}{(z_0+x)^2}$$ (since z_0 is const, $z = x$)

$$\therefore \quad mx = mg - \frac{k(i_0^2 + 2i_0u + u^2)}{(z_0+x)^2}$$

Using Taylor's equation to linearize,

Factor out z_0;

$$\frac{1}{(z_0+x^2)} = \frac{1}{z_0^2\left(1+\frac{x}{z_0}\right)^2}$$

Taylor Approximation:

$$(1+\varepsilon)^{-2} = 1-2\varepsilon$$

$$\therefore \varepsilon = \frac{x}{z_0}$$

$$\therefore \frac{1}{\left(1+\frac{x}{z_0}\right)^2} = 1-\frac{2x}{z_0}$$

$$\therefore \frac{1}{(z_0+x)^2} = \frac{1}{z_0^2}\left(1-\frac{2x}{z_0}\right)$$

$$\therefore \cancel{F_m = \frac{ki_0^2}{z_0^2} + \frac{2ki_0}{z_0^2}u} \quad mx = mg - \frac{ki_0^2}{z_0^2} - \frac{2ki_0}{z_0^2}u + \frac{2ki_0^2}{z_0^3}x$$

$$\therefore x = -\frac{2ki_0}{z_0^2}u + \frac{2ki_0^2}{z_0^3}x \qquad \left(\because mg = \frac{ki_0^2}{z_0^2}\right)$$

$$\therefore m\ddot{x} - \frac{2ki_0^2}{z_0^3}x = 0 \quad \& \quad ki_0^2 = mgz_0^2 \text{ using equilibrium}$$

$$\therefore \cancel{x = 2g} \quad \ddot{x} - \frac{2g}{z_0}x = 0$$

Compared with standard form, $\ddot{x} + \omega_n^2 x = 0$

With feedback, $u = k_p x + k_d \dot{x}$

$$m\ddot{x} - \frac{2ki_0^2}{z_0^3}x = -\frac{2ki_0^2 k_p}{z_0^2}x - \frac{2ki_0 k_d}{z_0^2}\dot{x}$$

$$c = \frac{2ki_0 k_d}{z_0^2}, \quad k_e = \frac{2ki_0 k_p}{z_0^2} - \frac{2ki_0^2}{z_0^3}$$

$$\therefore m\ddot{x} + c\dot{x} + k_e x = 0$$

Dividing by m

$$\therefore \ddot{x} + \frac{c}{m}\dot{x} + \frac{k_e}{m}x = 0$$

Matlab Code

```
% Parameters
m = 500;
c = 2000;
ke = 80000;

% Initial conditions
x_init = 0.01;
v_init = 0;
X0 = [x_init; v_init];
tspan = [0 5];

[t, X] = ode45(@(t,X) maglev_ode(t, X, m, c, ke), tspan, X0);

% Extract results
x = X(:,1);
v = X(:,2);

figure;
plot(t, x, 'LineWidth', 2);
xlabel('Time (s)');
ylabel('Displacement x (m)');
title('Suspension Response vs Displacement');
grid on

figure;
plot(t, v, 'LineWidth', 2);
xlabel('Time (s)');
ylabel('Velocity v (m/s)');
title('Suspension Response vs Velocity');
grid on

figure;
plot(x, v, 'LineWidth', 2);
xlabel('Displacement x (m)');
ylabel('Velocity v (m/s)');
title('Phase Plane Plot');
grid on;

function dXdt = maglev_ode(~, X, m, c, ke)
    dXdt = zeros(2,1);
    dXdt(1) = X(2);
    dXdt(2) = -(c/m)*X(2) - (ke/m)*X(1);
end
```

Result

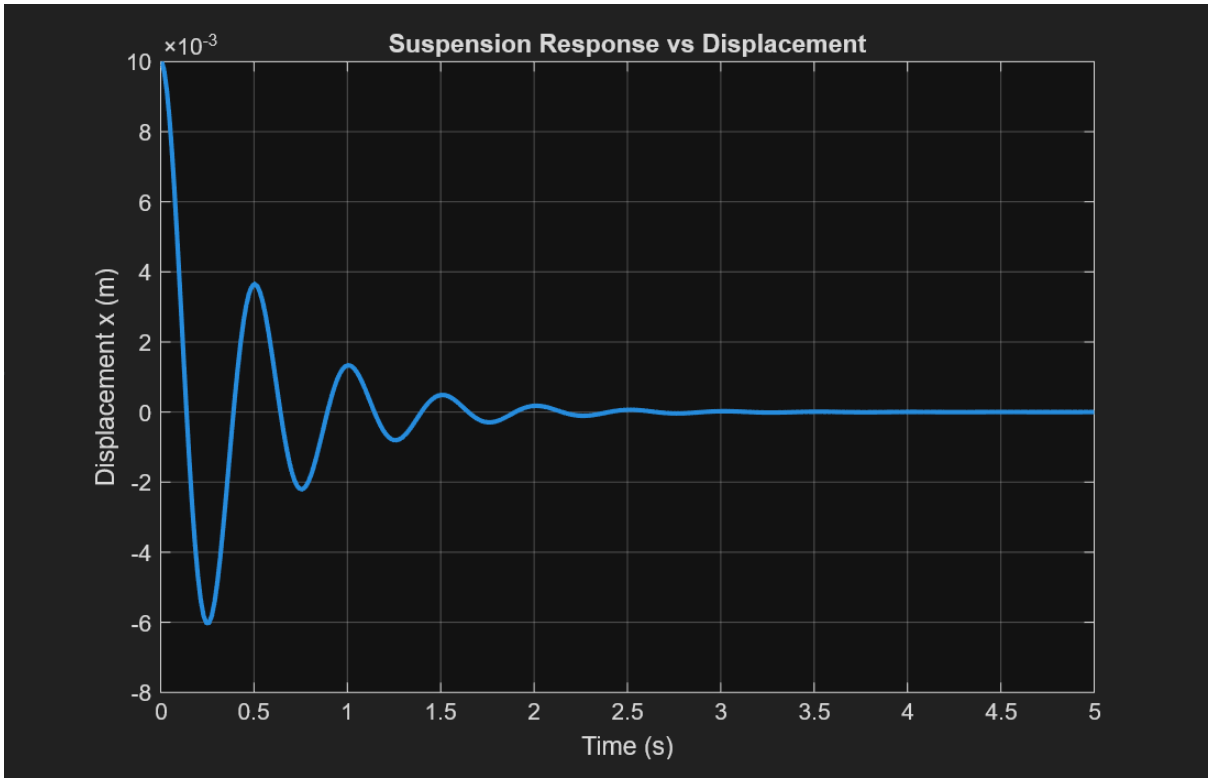


Figure 1 Displacement vs Time graph

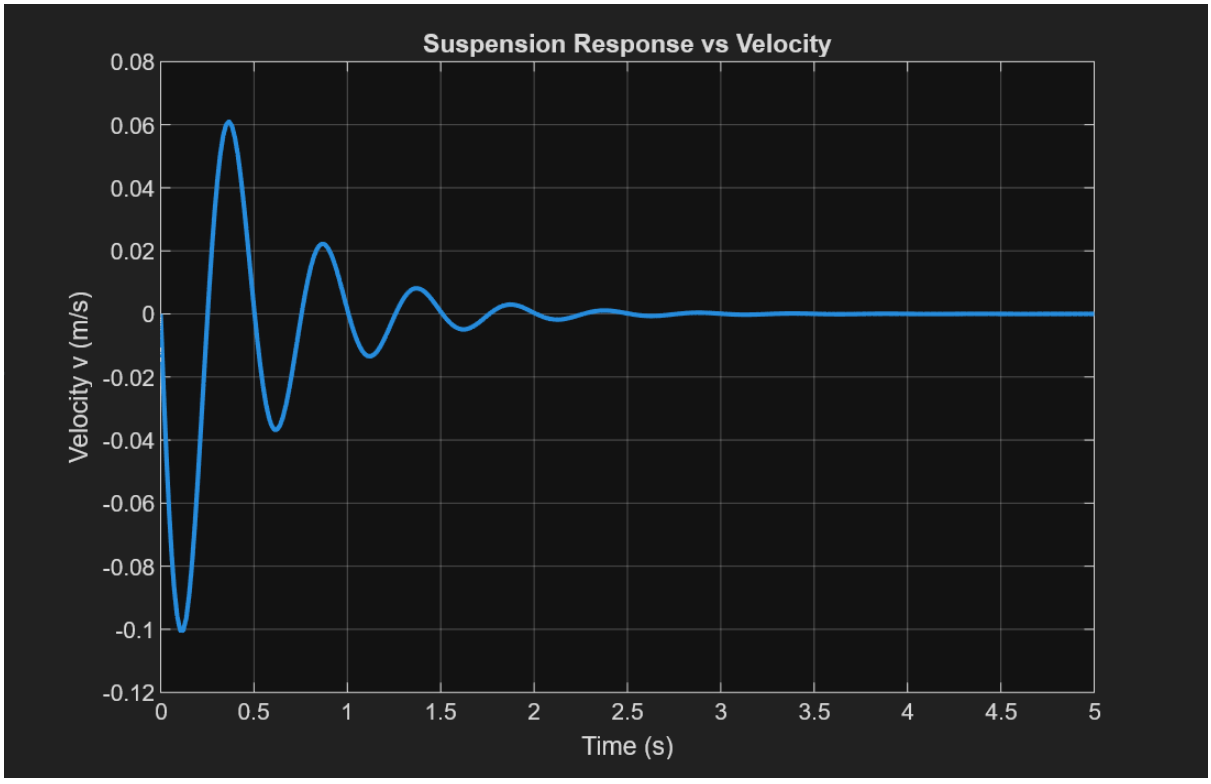


Figure 2 Velocity vs Time graph

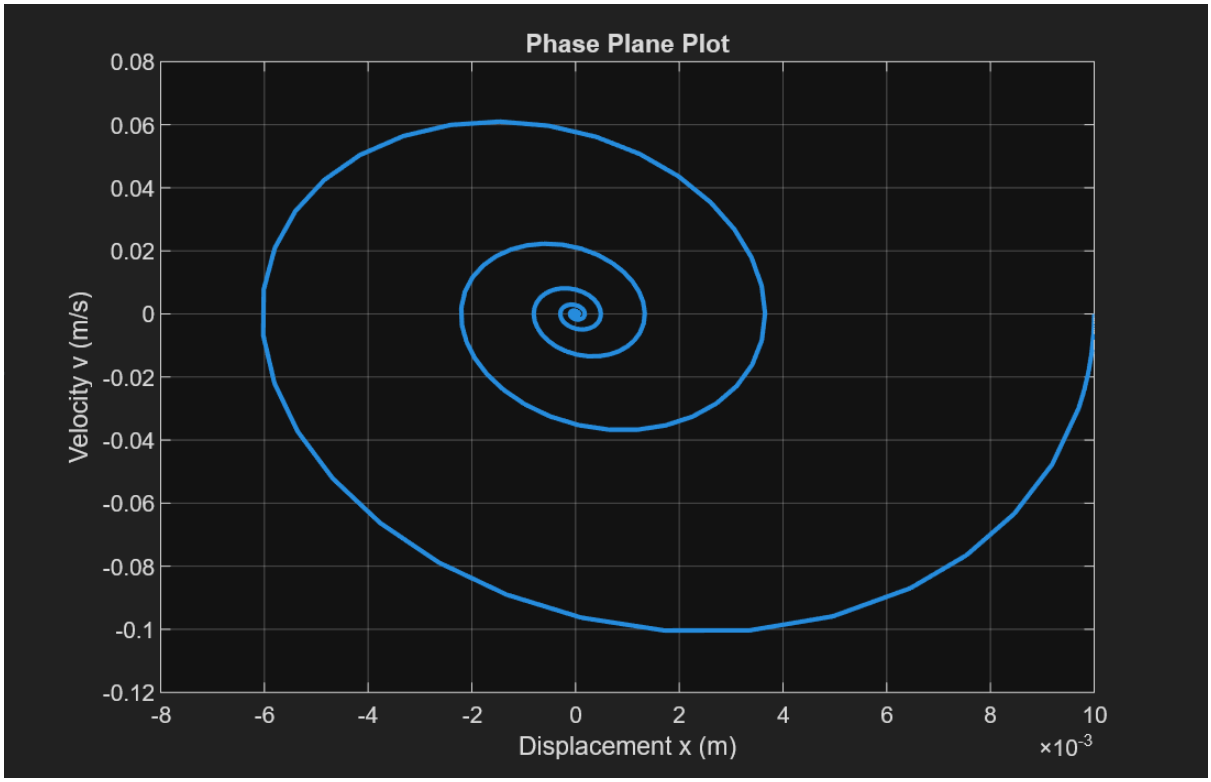


Figure 3 Velocity vs Displacement